

ANTONIUS' COMPENDIUM

Compendium of Knowledge Volume IV: Molecular Menagerie Version 0.001

Compiled by: Antonius W. Torode¹

1. Applied Research Laboratories - University Of Texas: Austin

Written in: L^AT_EX



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The original maintainer of this work is: Antonius Torode.

The current maintainer of this work is: Antonius Torode.

Chief Editor: Antonius Torode

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Preface

This document is a compilation of ideas, scratch work, derivations, useful formulas, definitions, constants, and general information used for my own studies as a reference while furthering self education. These are my notes. It's purpose is to provide a complete 'compendium' per say of various ideas used often. All the material in this document was either directly copied from one of the references listed at the end or derived from scratch. On occasion *typos may exist* due to human error but will be corrected when discovered.

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Topics Covered In This Book

- | | |
|----------------------------|------------------------|
| • Chemistry | • Pharmacology |
| • Physical Sciences | • Material Sciences |
| • Condensed Matter Physics | • Metallurgy |
| • Alchemy | • Chemical Engineering |
| • Homeopathy | • Toxicology |

The information in this book is in no way limited to the topics listed above. They serve as a simple guideline to what you will find within this document. For more information about this book or details about how to obtain your own copy please visit:

<https://torodean.github.io/>

“What we observe is not nature itself, but nature exposed to our method of questioning.” -
Heisenberg

Disclaimer

This book contains formulas, definitions, ideas and theorems that by nature are very precise. Due to this, some of the material in this book was taken directly from other sources such as but not limited to Wolfram Mathworld. This is only such in cases where a change in wording could cause ambiguities or loss of information quality. Following this, all sources used are listed in the references section and cited when used.

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(Yes, this is a contradiction.)

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Chemical Equivalences

The International Union of Pure and Applied Chemistry (IUPAC) is the universally-recognized authority on chemical nomenclature and terminology (according to themselves [1]).

1.1 Equivalences

IUPAC Name	Common Name	Formula	Where You Encounter It
Acetic Acid	Vinegar	$\text{CH}_3\text{CO}_2\text{H}$	Pickling, cleaning, cooking
Sodium Bicarbonate	Baking Soda	NaHCO_3	Baking, deodorizing, antacids
Sodium Hypochlorite	Bleach	NaOCl	Laundry and disinfection

Radiation

2.1 Introduction

Radiation is the emission and propagation of energy through space or matter. It is a fundamental phenomenon of nature, occurring across an enormous range of scales, from the transmission of radio signals and visible light to the energetic particles produced by nuclear reactions and cosmic events. Radiation serves as one of the primary means by which energy is transferred throughout the universe.

Radiation may take the form of electromagnetic waves, such as radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays, and gamma rays, or it may consist of particles, including electrons, protons, neutrons, and atomic nuclei. Regardless of its form, radiation is characterized by its ability to transport energy and momentum away from its source.

The interaction of radiation with matter governs countless natural and technological processes. These interactions determine how stars shine, how living organisms perceive light, how materials are analyzed, and how energy can be generated, transmitted, and measured. Depending on its energy and properties, radiation may pass through matter, be absorbed, scattered, or induce physical and chemical changes within the material it encounters.

The study of radiation lies at the intersection of physics, chemistry, biology, astronomy, and engineering. Understanding its origins, behavior, and effects is essential for fields ranging from medical imaging and radiation therapy to telecommunications, nuclear science, environmental monitoring, and space exploration.

This chapter examines the topic of radiation, which will end up being as broad or as focused as the information I happen to be studying at the time and where that leads me.

2.2 Discovery

Modern history records that Henri Becquerel first discovered **radioactivity** [3] at the turn of the 20th century.

Definition 2.2.1

Radioactivity is “the property possessed by some elements (such as uranium) or isotopes (such as carbon 14) of spontaneously emitting energetic particles (such as electrons or alpha particles) by the disintegration of their atomic nuclei [4].”

Early work with radioactivity involved experiments that showed some radiation could pass through aluminum foil - while others couldn't. It was determined by these that there were three basic types of radiation, which were named alpha, beta, and gamma radiation. A more modern and current understanding is that the **alpha particle** (α) is a helium nucleus (two protons and two neutrons), a **beta particle** (β) is a high speed electron, and **gamma rays** (γ) are high energy forms of electromagnetic waves (light).

General Chemistry

References

- [1] The International Union of Pure and Applied Chemistry <https://iupac.org/what-we-do/>
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- [3] Strom, Alexandra. Advanced General Chemistry, Smith College, 2024. <https://scholarworks.smith.edu/cgi/viewcontent.cgi?article=1006&context=textbooks>
- [4] Merriam Webster Dictionary <https://www.merriam-webster.com/dictionary>